

Paper G-03: AI Emergence: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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Shared Part IV notation and evidence boundaries are defined in `PART_IV_METHODS_AND_CLAIM_BOUNDARY.md`. This paper states only its local observables, comparison, and failure conditions.

Abstract

This paper develops a falsifiable CDFD/AFL model of AI Emergence. It maps data, compute, models, feedback, deployment demand, and social attention to driving flux Φ , compute, data quality, governance, evaluation, energy, and oversight to active constraint C , learning, tool use, modularity, monitoring, correction, and institutional response to responsiveness S , and weights, deployed systems, standards, incidents, incentives, and feedback history to structural memory M_s . The target outcome is capability, reliability, diffusion, risk, and social adaptation, tested against scaling records, evaluations, deployment telemetry, incidents, policy changes, and energy use. The paper treats universality as a shared model architecture rather than an assertion that unlike systems have the same material mechanism. It states the negative result that would make the mapping unnecessary and places the domain claim inside the cumulative CDFD programme and its reproducible run-time boundary.

1 Introduction

Artificial neural networks simulate learning by iteratively adjusting millions or billions of parameters to minimize a loss function against a training dataset. The current paradigm of AI development relies on massive brute-force scaling: pumping enormous volumes of data through immense clusters of GPUs.

While this has yielded emergent cognitive behaviors, the trajectory is fundamentally thermodynamic. The network is essentially an active engine consuming information and energy to produce structured mathematical memory. In CDFD terms, a deep learning model is an adaptive topological surface attempting to balance immense throughput against strict physical and architectural bottlenecks.

2 Neural Architecture as an Adaptive Surface

We map the training of a neural network to the CDFD life number equation:

$$\Psi_s = \left(\frac{\Phi}{C}\right) \cdot S \cdot M_s \quad (1)$$

In an artificial intelligence system:

- **Flux** (Φ): The gradient updates and the raw throughput of training tokens (data flow).
- **Constraint** (C): The rigid, fixed architecture of the network (e.g., fixed number of layers and attention heads), as well as the physical hardware limits (memory bandwidth and thermal dissipation).
- **Responsiveness** (S): The available parameter space (model width and depth) that can actively adapt to the incoming gradients.
- **Memory** (M_s): The accumulated, frozen weights of the network. As the model trains, it transitions from a high-entropy chaotic state into a low-entropy locked state (overfitting or saturation).

2.1 The Saturation Threshold and Catastrophic Forgetting

During early training, S is high (many free parameters) and M_s is low. The network efficiently absorbs the flux Φ . $\Psi_s \approx 1$.

However, as training progresses, the weights solidify into fixed representations. The systemic memory M_s increases, locking the topology. If new, out-of-distribution data (Φ) is introduced, the fixed architectural constraints (C) cannot flex to accommodate it. The network enters an over-constrained regime ($\Psi_s \ll 1$). To learn the new data, it must overwrite its existing weights, destroying its previous capabilities—a phenomenon known in machine learning as "catastrophic forgetting."

Unlike biological brains, which dynamically grow new synapses and prune old ones (active restructuring of C and S), current LLMs have fixed, static constraints.

3 The Path to General Intelligence

The CDFD framework models the condition that that you cannot achieve infinite adaptability in a system with fixed static constraints. If C is locked, Ψ_s will enter an overload regime as M_s saturates.

True Artificial General Intelligence (AGI) requires an architecture where the constraint C itself is a dynamic variable governed by the differential equation $\frac{\partial C}{\partial t} = \alpha|\Phi| - \beta C$. The network must be able to physically or logically grow new architectural branches (neurogenesis) and prune useless nodes to continually maintain $\Psi_s \approx 1$. Liquid neural networks and sparsely activated mixture-of-experts (MoE) are primitive steps toward this, but an AGI must be a fully self-regulating CDFD surface.

4 Measurement Layer

5 Cross-Domain Model Upgrade: Mechanism, Scale, and Tests

5.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For AI Emergence, the proposed drive is data, compute, models, feedback, deployment demand, and social attention. The active limitation is compute, data quality, governance, evaluation, energy, and oversight. Responsiveness is carried by learning, tool use, modularity, monitoring, correction, and institutional response, while retained state is represented by weights, deployed systems, standards, incidents, incentives, and feedback history. The outcome to explain is capability, reliability, diffusion, risk, and social adaptation.

Table 1: Operational CDFD/AFL map for Paper G-03.

Term	Paper-specific observable meaning	Measurement question
Φ	data, compute, models, feedback, deployment demand, and social attention	What rate, load, gradient, or demand enters the focal system?
C	compute, data quality, governance, evaluation, energy, and oversight	Which measured bottleneck limits transfer, function, or recovery?
S	learning, tool use, modularity, monitoring, correction, and institutional response	Which process changes effective capacity or reroutes the load?
M_s	weights, deployed systems, standards, incidents, incentives, and feedback history	Which retained state makes equal present loads produce different futures?
Y	capability, reliability, diffusion, risk, and social adaptation	Which outcome is predicted out of sample, with uncertainty?

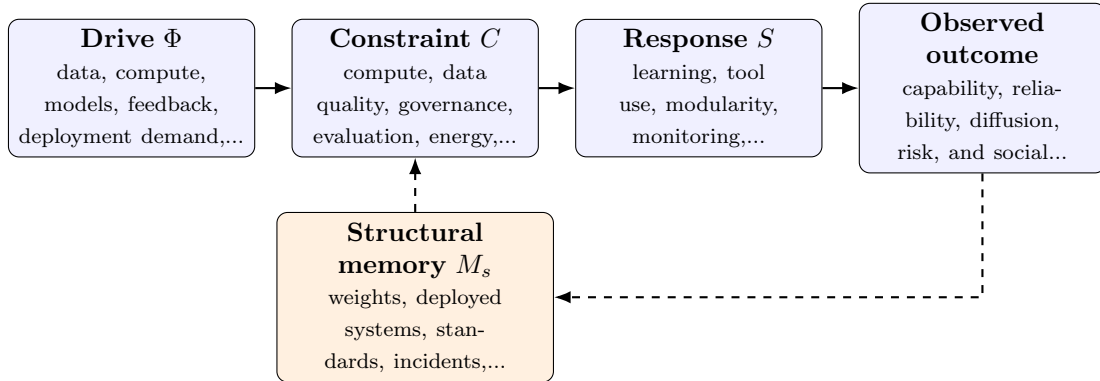


Figure 1: Paper G-03 observable CDFD/AFL architecture for AI Emergence. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

5.2 Paper-Specific Causal Hypothesis

For this paper, the hypothesized causal sequence is specific. A change in data, compute, models, feedback, deployment demand, and social attention arrives faster than learning, tool use, modularity, monitoring, correction, and institutional response can compensate. This raises or redistributes compute, data quality, governance, evaluation, energy, and oversight. If the load

persists, the retained state encoded by weights, deployed systems, standards, incidents, incentives, and feedback history changes the next response, so a later event of equal magnitude can produce a different trajectory in capability, reliability, diffusion, risk, and social adaptation. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

5.3 Cross-Scale Translation Without Mechanistic Erasure

For the abstract and cognitive systems family, the cross-domain translation compares information-bearing activity, limited channels or institutions, adaptive reorganization, and retained semantic or technical structure. Because meanings and goals are not conserved physical quantities, every mapping must state its observational unit and avoid treating metaphor as measurement. [3, 2, 5, 4, 1] The required scale declaration for this paper is microseconds to decades; model components to socio-technical ecosystems. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

5.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper G-03. Each rung can fail independently.

Test	Design	Discriminating result
Proxy audit	Measure scaling records, evaluations, deployment telemetry, incidents, policy changes, and energy use and preregister how each record maps to Φ , C , S , M_s , and Y .	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a capability or deployment shock under contrasting oversight histories while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper’s stated falsifier is: CDFD variables do not improve capability, failure, or governance-response prediction.

5.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from scaling records, evaluations, deployment telemetry, incidents, policy changes, and energy use and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state

extension and test whether it improves prediction of capability, reliability, diffusion, risk, and social adaptation. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the cross-domain translation audit.

6 Conclusion

Artificial-intelligence systems face measurable limits in compute, data, energy, evaluation, and governance. CDFD offers a candidate architecture for testing how those constraints interact with learning and deployment history; it does not establish a universal ceiling or guarantee open-ended cognition.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part’s `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

7 Reference Layer

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